Spectral Perturbations and Eigengap Sensitivity

Spectral Stability of Equity Covariance

Mathematical Foundations of Data Science
University of Vienna

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Abstract

Population covariance matrices are inferred from finite samples, so their empirical spectra need not agree with population spectra. We first derive the finite-sample bias of the mean-centered covariance estimator and its usual unbiased correction. We then analyze a symmetric two-dimensional perturbation through directional variance, obtaining the rotation of its leading eigenvector in terms of the perturbation and population eigengap. Finally, a Rayleigh-quotient argument bounds the displacement of the leading eigenvalue and contrasting examples show why eigenvalue stability does not, by itself, imply stability of eigenvectors.

1 Population and sample covariance

Let $R_1, \dots, R_n \in \mathbb{R}^d$ be independent, identically distributed return vectors with finite second moments, where $n \geq 2$. Write

$$\mu = \mathbb{E}[R_t], \qquad \Sigma = \mathrm{Cov}(R_t) = \mathbb{E}[(R_t - \mu)(R_t - \mu)^\mathsf{T}], \qquad \overline{R} = \frac{1}{n}\sum_{t=1}^{n} R_t. \tag{1}$$

The mean-centered covariance matrix with denominator n is

$$\widehat{\Sigma}_n = \frac{1}{n}\sum_{t=1}^{n}(R_t - \overline{R})(R_t - \overline{R})^\mathsf{T}. \tag{2}$$

Its dependence on the estimated mean changes its expectation.

Lemma 1.1 (Centering identity)**.** *For any vectors $R_1, \dots, R_n$ and any $\mu \in \mathbb{R}^d$,*

$$\sum_{t=1}^{n}(R_t - \overline{R})(R_t - \overline{R})^\mathsf{T} = \sum_{t=1}^{n}(R_t - \mu)(R_t - \mu)^\mathsf{T} - n(\overline{R} - \mu)(\overline{R} - \mu)^\mathsf{T}. \tag{3}$$

Proof. Since $R_t - \overline{R} = (R_t - \mu) - (\overline{R} - \mu)$, expanding the outer products gives

$$\begin{aligned}
&\sum_{t=1}^{n}(R_t - \overline{R})(R_t - \overline{R})^\mathsf{T} \\
&\quad = \sum_{t=1}^{n}(R_t - \mu)(R_t - \mu)^\mathsf{T} - \sum_{t=1}^{n}(R_t - \mu)(\overline{R} - \mu)^\mathsf{T} \\
&\qquad - (\overline{R} - \mu)\sum_{t=1}^{n}(R_t - \mu)^\mathsf{T} + n(\overline{R} - \mu)(\overline{R} - \mu)^\mathsf{T}.
\end{aligned}$$

Using $\sum_{t=1}^{n}(R_t - \mu) = n(\overline{R} - \mu)$, the two cross terms combine with the last term to yield (3). □

The subtraction in (3) records the effect of estimating the mean rather than centering about the population mean.

Proposition 1.2 (Bias and unbiased correction)**.** *Under the assumptions of* (1)*,*

$$\mathbb{E}[\widehat{\Sigma}_n] = \frac{n-1}{n}\Sigma. \tag{4}$$

Consequently the corrected estimator

$$S = \frac{1}{n-1}\sum_{t=1}^{n}(R_t - \overline{R})(R_t - \overline{R})^{\mathsf{T}} \tag{5}$$

satisfies $\mathbb{E}[S] = \Sigma$*.*

Proof. Taking expectations in (3) and dividing by n gives

$$\mathbb{E}[\widehat{\Sigma}_n] = \Sigma - \mathbb{E}[(\overline{R} - \mu)(\overline{R} - \mu)^{\mathsf{T}}]. \tag{6}$$

To calculate the second term, write

$$\mathbb{E}[(\overline{R} - \mu)(\overline{R} - \mu)^{\mathsf{T}}] = \frac{1}{n^2}\sum_{t=1}^{n}\sum_{s=1}^{n}\mathbb{E}[(R_t - \mu)(R_s - \mu)^{\mathsf{T}}].$$

The n diagonal terms are equal to Σ. For $t \neq s$, independence and $\mathbb{E}[R_t - \mu] = 0$ make the cross terms zero. Therefore

$$\mathbb{E}[(\overline{R} - \mu)(\overline{R} - \mu)^{\mathsf{T}}] = \frac{1}{n}\Sigma.$$

Substitution into (6) proves (4); multiplying by $n/(n-1)$ gives $\mathbb{E}[S] = \Sigma$. □

Unbiasedness concerns the expectation of the matrix. It does not imply that the eigenvectors of an individual sample covariance matrix coincide with those of Σ.

2 A two-dimensional eigenvector perturbation

Consider the population covariance matrix and a symmetric perturbation

$$\Sigma = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}, \qquad E = \begin{pmatrix}0 & \varepsilon\\ \varepsilon & 0\end{pmatrix}, \qquad \widehat{\Sigma} = \Sigma + E, \tag{7}$$

where $\lambda_1 > \lambda_2 > 0$ and $\varepsilon > 0$. For $\widehat{\Sigma}$ to remain a covariance matrix it is necessary and sufficient that $\varepsilon^2 \leq \lambda_1\lambda_2$; the angular calculation itself only needs symmetry. The leading population eigenvector is $e_1 = (1,0)^{\mathsf{T}}$, and we seek the leading unit eigenvector of $\widehat{\Sigma}$.

Proposition 2.1 (Leading-eigenvector rotation)**.** *Let* $\delta = \lambda_1 - \lambda_2 > 0$*. The leading unit eigenvector of* $\widehat{\Sigma}$*, with positive first coordinate, has the form*

$$\widehat{v}_1 = \begin{pmatrix}\cos\theta_*\\ \sin\theta_*\end{pmatrix}, \qquad \theta_* = \frac{1}{2}\arctan\left(\frac{2\varepsilon}{\delta}\right) \in \left(0, \frac{\pi}{4}\right). \tag{8}$$

Proof. For a unit vector $u(\theta) = (\cos\theta, \sin\theta)^{\mathsf{T}}$, directional variance is

$$\begin{aligned} f(\theta) = u(\theta)^{\mathsf{T}}\widehat{\Sigma}u(\theta) &= \lambda_1\cos^2\theta + 2\varepsilon\cos\theta\sin\theta + \lambda_2\sin^2\theta \\ &= \frac{\lambda_1 + \lambda_2}{2} + \frac{\delta}{2}\cos 2\theta + \varepsilon\sin 2\theta. \end{aligned} \tag{9}$$

Differentiating gives

$$f'(\theta) = -\delta\sin 2\theta + 2\varepsilon\cos 2\theta. \tag{10}$$

At the maximizing angle in $[0, \pi/2]$, the derivative vanishes, so

$$\tan 2\theta_* = \frac{2\varepsilon}{\delta}. \tag{11}$$

Because $\delta, \varepsilon > 0$, this determines $0 < \theta_* < \pi/4$. Moreover,

$$\cos 2\theta_* = \frac{\delta}{\sqrt{\delta^2 + 4\varepsilon^2}}, \qquad \sin 2\theta_* = \frac{2\varepsilon}{\sqrt{\delta^2 + 4\varepsilon^2}},$$

and hence

$$f''(\theta_*) = -2\delta \cos 2\theta_* - 4\varepsilon \sin 2\theta_* = -2\sqrt{\delta^2 + 4\varepsilon^2} < 0. \tag{12}$$

Finally, the trigonometric form of (9) is maximized globally when $(\cos 2\theta, \sin 2\theta)$ points in the direction of $(\delta, 2\varepsilon)$. Thus $u(\theta_*)$ maximizes directional variance and is the leading eigenvector. □

The corresponding eigenvalues follow directly from the characteristic polynomial:

$$\det(\widehat{\Sigma} - \eta I_2) = (\lambda_1 - \eta)(\lambda_2 - \eta) - \varepsilon^2, \tag{13}$$

$$\eta_\pm = \frac{\lambda_1 + \lambda_2 \pm \sqrt{(\lambda_1 - \lambda_2)^2 + 4\varepsilon^2}}{2}. \tag{14}$$

In particular, $\eta_+ \geq \eta_-$ and η_+ is associated with $\widehat{v}_1$.

Remark 2.2 (Perturbation relative to the eigengap). Equation (8) makes the ratio ε/δ decisive for rotation in this example. If ε/δ is small, then the first-order expansion of arctan yields

$$\theta_* \sim \frac{\varepsilon}{\delta} \qquad \text{as } \frac{\varepsilon}{\delta} \longrightarrow 0. \tag{15}$$

A small absolute perturbation does not by itself ensure a small rotation; the spectral separation matters.

2.1 A vanishing-perturbation example with nonvanishing rotation

Let $c > 0$ be fixed and let $t > 0$. Consider

$$\Sigma(t) = \begin{pmatrix} 1+t & 0 \\ 0 & 1 \end{pmatrix}, \qquad E(t) = \begin{pmatrix} 0 & ct \\ ct & 0 \end{pmatrix}, \qquad \widehat{\Sigma}(t) = \begin{pmatrix} 1+t & ct \\ ct & 1 \end{pmatrix}. \tag{16}$$

Here $\delta(t) = t$, $\varepsilon(t) = ct$, and $\|E(t)\|_2 = ct \to 0$, while

$$\theta_*(t) = \frac{1}{2}\arctan(2c), \tag{17}$$

which does not tend to zero. Meanwhile, the perturbed eigenvalues

$$\eta_\pm(t) = \frac{2 + t \pm \sqrt{t^2 + 4c^2t^2}}{2} \tag{18}$$

converge to 1, as do the two population eigenvalues. In particular,

$$|\eta_+(t) - (1+t)| = \frac{\sqrt{t^2 + 4c^2t^2} - t}{2} \longrightarrow 0,$$

and likewise $|\eta_-(t) - 1| \to 0$. Thus the eigenvalues become arbitrarily close while the leading eigendirections retain a fixed nonzero angle. For sufficiently small t, both matrices in (16) are positive definite.

3 The Rayleigh quotient and the leading eigenvalue

For a symmetric matrix $A \in \mathbb{R}^{d\times d}$ and $x \neq 0$, define its Rayleigh quotient by

$$\mathcal{R}_A(x) = \frac{x^\mathsf{T} A x}{x^\mathsf{T} x}. \tag{19}$$

For a unit vector, this is simply $x^\mathsf{T} A x$. The variational characterization below explains the earlier maximization of directional variance.

Proposition 3.1 (Largest eigenvalue as maximal directional variance)**.** *If $A = A^{\mathsf{T}} \in \mathbb{R}^{d\times d}$ has eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$, then*

$$\lambda_1 = \max_{\|x\|_2=1} x^{\mathsf{T}} A x. \tag{20}$$

Proof. Let $v_1, \ldots, v_d$ be an orthonormal eigenbasis, and write any unit vector as $x = \sum_{i=1}^{d} a_i v_i$ with $\sum_i a_i^2 = 1$. Then

$$x^{\mathsf{T}} A x = \sum_{i=1}^{d} \lambda_i a_i^2 \leq \lambda_1 \sum_{i=1}^{d} a_i^2 = \lambda_1.$$

Equality holds for $x = v_1$, proving the claim. □

For example, for $A = \mathrm{diag}(6, 2)$ and $x = 2^{-1/2}(1, 1)^{\mathsf{T}}$, one has

$$\mathcal{R}_A(x) = x^{\mathsf{T}} A x = \frac{1}{2}(6 + 2) = 4,$$

which is not an eigenvalue. The Rayleigh quotient measures variance in an arbitrary unit direction, not necessarily an eigendirection.

Proposition 3.2 (Perturbation bound for the largest eigenvalue)**.** *Let $A, E \in \mathbb{R}^{d\times d}$ be symmetric, and set $\widehat{A} = A + E$. Then*

$$|\lambda_1(\widehat{A}) - \lambda_1(A)| \leq \|E\|_2. \tag{21}$$

Proof. For any unit vector x,

$$x^{\mathsf{T}} \widehat{A} x = x^{\mathsf{T}} A x + x^{\mathsf{T}} E x \leq x^{\mathsf{T}} A x + \|E\|_2.$$

Maximizing over unit vectors and using (20) yields

$$\lambda_1(\widehat{A}) \leq \lambda_1(A) + \|E\|_2.$$

The identity $A = \widehat{A} - E$ gives the reverse inequality $\lambda_1(A) \leq \lambda_1(\widehat{A}) + \|E\|_2$. Combining them proves (21). □

This is the leading-eigenvalue case of Weyl's inequality. It controls eigenvalue changes through perturbation magnitude alone, without reference to the eigengap.

4 Eigenvalue versus eigenvector stability

Return to (7). Its leading eigenvector rotates by

$$\theta_* = \frac{1}{2} \arctan\left(\frac{2\varepsilon}{\delta}\right), \qquad \delta = \lambda_1 - \lambda_2 > 0.$$

For fixed $\varepsilon > 0$, letting $\delta \downarrow 0$ gives $\theta_* \to \pi/4$: a rotation of 45°. More strongly, this phenomenon can occur with perturbations that also tend to zero.

Set, for $t > 0$,

$$\Sigma(t) = \begin{pmatrix} 1 + t^2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad E(t) = \begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}. \tag{22}$$

The perturbation magnitude satisfies $\|E(t)\|_2 = t \to 0$, while the eigengap is $\delta(t) = t^2$. Therefore

$$\tan(2\theta_*(t)) = \frac{2t}{t^2} = \frac{2}{t} \longrightarrow \infty, \qquad \text{and hence} \qquad \theta_*(t) \longrightarrow \frac{\pi}{4}. \tag{23}$$

Both $\Sigma(t)$ and $\Sigma(t) + E(t)$ are positive definite for $t > 0$, since the latter has determinant 1. By (21), their leading eigenvalues still differ by at most t, which tends to zero.

These examples separate two phenomena. Small operator-norm errors control eigenvalue errors, but control of eigenvector rotation requires information about the eigengap relative to the perturbation. The direction of the perturbation also matters: the off-diagonal perturbations considered here explicitly couple the two eigendirections. When the population eigenvalues coalesce in the limit, the individual leading direction is no longer uniquely determined.